

1.

a) 连接student、takes、course表再指定要求即可

```
select distinct student.ID, student.name
from student,takes,course
where student.ID = takes.ID and course.course_id = takes.course_id and
course.dept_name = 'Comp.Sci'
```

b) 先查找选修过2017年之前的，减去这个结果集即可

```
select student.ID, student.name
from student
where not exists (
select 1
from takes
where takes.ID = student.ID and takes.year < 2017
)
```

c) 利用集合函数max

```
select dept_name, max(salary) as max_salary
from instructor
group by dept_name
```

d) 利用c)结果和集合函数min

```
with dept_max_salary(dept_name,max_salary) as
(select dept_name, max(salary) as max_salary
from instructor
group by dept_name
)
select min(max_salary)
from dept_max_salary
```

2. 改写如下:

```
select dept_name
from (
select dept_name, sum(salary) as value
from instructor
```

```

group by dept_name
)
where value > (
select avg(value)
from (
        select sum(salary) as value
        from instructor
        group by dept_name
    )
)

```

3. 改写如下，利用集合函数count来找到所有含重复的title的结果集，并对该结果集是否为空进行判断，如果为空则所有行都不重复，返回true，否则返回false：

```

where not exists
(
select title
from course
group by title
having count(*) > 1
)

```

4. a) 利用关键字in来实现

```

select ID
from depositor
where ID not in (
    select ID
    from borrower
)

```

b)

```

select A.ID
from customer as A, customer as B
where B.customer_name = '12345' and B.customer_street = A.customer_street and
B.customer_city = A.customer_city and A.customer_name <> '12345'

```

c)利用join操作即可

```
select distinct account.branch_name
from account
join depositor on account.account_number = depositor.account_number
join customer on depositor.ID = customer.ID
where customer.customer_city = 'Harrison';
```

5. a) 思路是找出每一个用户的所有分行账户，用brooklyn所有分行去与该用户所有分行作“差集”，并通过判断是否为空来实现目的，若为空，则说明该用户有brooklyn所有分行的账户，是所要找目标客户，否则不是。

```
select distinct d.ID
from depositor d
where not exists (
(
select branch_name
from branch
where branch.branch_city = 'Brooklyn'
)
except
(
select a.branch_name
from depositor d2 join account a on d2.account_number = a.account_number
where d.ID = d2.ID
)
)
```

b)

```
select sum(amount)
from loan;
```

c) 利用some即可

```
select b.branch_name
from branch b
where b.assets > some (
select b2.assets
from branch b2
where b2.city = 'Brooklyn'
)
```